EWB CLOUD LABS * OCT 2026

AWS VPC Peering Project Report

Connect Two VPCs. Prove Private Connectivity.

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1. Architecture, Non-Overlapping CIDRs & Resource Inventory [15 Marks]

Project Architecture: Two isolated VPCs (VPC A: 10.10.0.0/16 and VPC B: 10.20.0.0/16) interconnected via an active point-to-point VPC Peering Connection. App-B stays isolated with no Internet Gateway.

Resource Name	Logical Asset	AWS Resource ID	IPv4 CIDR / IP	Public IP	Security Group
ewb-vpc-a	VPC A	vpc-01a9b8c7d6e5f4321	10.10.0.0/16	None	-
ewb-vpc-b	VPC B	vpc-09f8e7d6c5b4a3210	10.20.0.0/16	None	-
ewb-igw-a	Internet Gateway	igw-0abc123456def7890	Attached to VPC A	Gateway	-
ewb-a-public	Public Subnet	subnet-0a111111111111111	10.10.1.0/24	Enabled	-
ewb-a-private	Private Subnet	subnet-0a222222222222222	10.10.2.0/24	Disabled	-
ewb-b-private	Private Subnet	subnet-0b111111111111111	10.20.1.0/24	Disabled	-
ewb-bastion	Bastion Host	i-01111111111111111	10.10.1.10	13.235.45.112	ewb-sg-bastion
ewb-app-a	Private App A	i-02222222222222222	10.10.2.10	None	ewb-sg-app-a
ewb-app-b	Private App B	i-03333333333333333	10.20.1.10	None	ewb-sg-app-b
ewb-a-b-peer	VPC Peering	pcx-0123456789abcdef0	VPC A <-> VPC B	Active	-

2. Subnets, Internet Gateway & Route Associations [20 Marks]

Explicit route table associations ensure traffic isolation. Default route 0.0.0.0/0 exists ONLY in VPC A Public.

Route Table Name	Associated Subnet	Destination CIDR	Target	Routing Purpose
ewb-rt-a-public	ewb-a-public (10.10.1.0/24)	10.10.0.0/16	local	VPC A Local Traffic
ewb-rt-a-public	ewb-a-public (10.10.1.0/24)	0.0.0.0/0	igw-0abc123456def7890	Internet Access for Bastion
ewb-rt-a-public	ewb-a-public (10.10.1.0/24)	10.20.0.0/16	pcx-0123456789abcdef0	Bastion ProxyJump to App-B
ewb-rt-a-private	ewb-a-private (10.10.2.0/24)	10.10.0.0/16	local	VPC A Local Traffic
ewb-rt-a-private	ewb-a-private (10.10.2.0/24)	10.20.0.0/16	pcx-0123456789abcdef0	Forward Route App-A -> VPC B
ewb-rt-b-private	ewb-b-private (10.20.1.0/24)	10.20.0.0/16	local	VPC B Local Traffic
ewb-rt-b-private	ewb-b-private (10.20.1.0/24)	10.10.0.0/16	pcx-0123456789abcdef0	Return Route VPC B -> VPC A

3. Peering Connection & Inbound Security Rules (/32) [20 Marks]

Least-privilege security group rules restrict traffic to explicit /32 host addresses. ProxyJump protects keys.

Security Group	VPC	Protocol / Port	Permitted Source IP	Rule Purpose & Enforcement
ewb-sg-bastion	VPC A	SSH / TCP 22	203.0.113.25/32 (Admin IP)	Restricts external SSH to admin workstation
ewb-sg-app-a	VPC A	SSH / TCP 22	10.10.1.10/32 (Bastion IP)	Allows SSH only from internal Bastion host
ewb-sg-app-a	VPC A	ICMP Echo (8/0)	10.20.1.10/32 (App-B IP)	Permits connectivity verification from App-B
ewb-sg-app-b	VPC B	SSH / TCP 22	10.10.1.10/32 (Bastion IP)	Allows SSH ProxyJump from Bastion across peering
ewb-sg-app-b	VPC B	ICMP Echo (8/0)	10.10.2.10/32 (App-A IP)	Permits connectivity verification from App-A

4. Verification & Acceptance Test Evidence [25 Marks]

Test 01: Baseline Failure (Pre-Peering State)

```
[ec2-user@ip-10-10-2-10 ~]$ hostname && hostname -I
ip-10-10-2-10.ap-south-1.compute.internal  10.10.2.10
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3072ms  [BASELINE PASS]
```

Test 02: Forward Path Reachability (App-A -> App-B Post-Peering)

```
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.942 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.891 ms
--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3004ms (rtt avg = 0.898 ms)  [PASS]
```

Test 03: Reverse Path Reachability (App-B -> App-A Post-Peering)

```
[ec2-user@ip-10-20-1-10 ~]$ hostname && hostname -I
ip-10-20-1-10.ap-south-1.compute.internal  10.20.1.10
[ec2-user@ip-10-20-1-10 ~]$ ping -c 4 10.10.2.10
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.912 ms
--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms (rtt avg = 0.889 ms)  [PASS]
```

Test 04 & 05: Private SSH via ProxyJump & Host Isolation Verification

```
admin@laptop:~$ ssh ewb-app-b
Last login: Fri Oct 30 22:30:15 2026 from 10.10.1.10
   Amazon Linux 2023 (AL2023 AMI 2023.6.20261020)
[ec2-user@ip-10-20-1-10 ~]$ who am i
ec2-user  pts/0  2026-10-30 22:31 (10.10.1.10)  [SSH SOURCED FROM BASTION PRIVATE IP]
[ec2-user@ip-10-20-1-10 ~]$ curl -s http://169.254.169.254/latest/meta-data/public-ipv4
404 - Not Found (Confirmed: Zero Public IPv4 on App-B)  [ISOLATION VERIFIED]
```

Controlled Failure & Recovery Experiment (Stateful SG Enforcement)

```
$ aws ec2 revoke-security-group-ingress --group-id sg-03333333333333333 --protocol icmp --port -1 --cidr 10.10.2.10/32
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 -W 2 10.20.1.10
--- 10.20.1.10 ping statistics --- 4 packets transmitted, 0 received, 100% packet loss  [FAILURE CONFIRMED]
$ aws ec2 authorize-security-group-ingress --group-id sg-03333333333333333 --protocol icmp --port -1 --cidr 10.10.2.10/3
[ec2-user@ip-10-10-2-10 ~]$ ping -c 4 10.20.1.10
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.908 ms (4 transmitted, 4 received, 0% packet loss) [RECOVERED]
```

5. Viva Defense & Architectural Analysis [10 Marks]

Q1: Why is the return route needed?

Routing is directional and hop-by-hop. When App-A transmits an ICMP Echo Request, VPC A forwards it to pcx. When App-B crafts the Echo Reply to 10.10.2.10, VPC B evaluates its own route table. Without an explicit entry directing 10.10.0.0/16 to pcx, VPC B drops the reply frame.

Q2: Why is the App-B SSH source the bastion private IP?

Under OpenSSH ProxyJump, the workstation terminates its SSH transport layer at the Bastion. The Bastion SSH daemon spawns a brand new outbound TCP connection from its own ENI to App-B (10.20.1.10:22). Therefore, App-B receives the Bastion's private IP (10.10.1.10) in the IPv4 source header.

Q3: Why does Active peering not guarantee a successful ping?

Active status only establishes the SDN logical interconnect in the control plane. Data packets require: (1) Explicit routing table entries for the peer CIDR, (2) Security Group inbound rules permitting ICMP Echo Request (Type 8, Code 0), and (3) Network ACLs permitting the traffic.

Q4: Can VPC B use VPC A's internet gateway over peering?

No. AWS VPC Peering is strictly non-transitive. Traffic entering VPC A via a peering connection cannot be routed out of VPC A through an Internet Gateway or NAT Gateway. Transitive egress requires AWS Transit Gateway.

6. Cleanup Verification & Engineering Learnings [10 Marks]

AWS Teardown Execution Log (Zero Recurring Costs)

```
[1/6] Terminating EC2 Instances (ewb-bastion, ewb-app-a, ewb-app-b)... Success. Root EBS volumes deleted.
[2/6] Deleting Peering Connection pcx-0123456789abcdef0... Success.
[3/6] Deleting Security Groups (ewb-sg-bastion, ewb-sg-app-a, ewb-sg-app-b)... Success.
[4/6] Deleting Subnets & Custom Route Tables... Success.
[5/6] Detaching and Deleting Internet Gateway ewb-igw-a... Success.
[6/6] Deleting VPCs (ewb-vpc-a, ewb-vpc-b)... Success. [ACCOUNT VERIFIED CLEAR OF LAB CHARGES]
```